Slides 06/01

Méthodes efficaces pour l'estimation de modèles spatio-temporel

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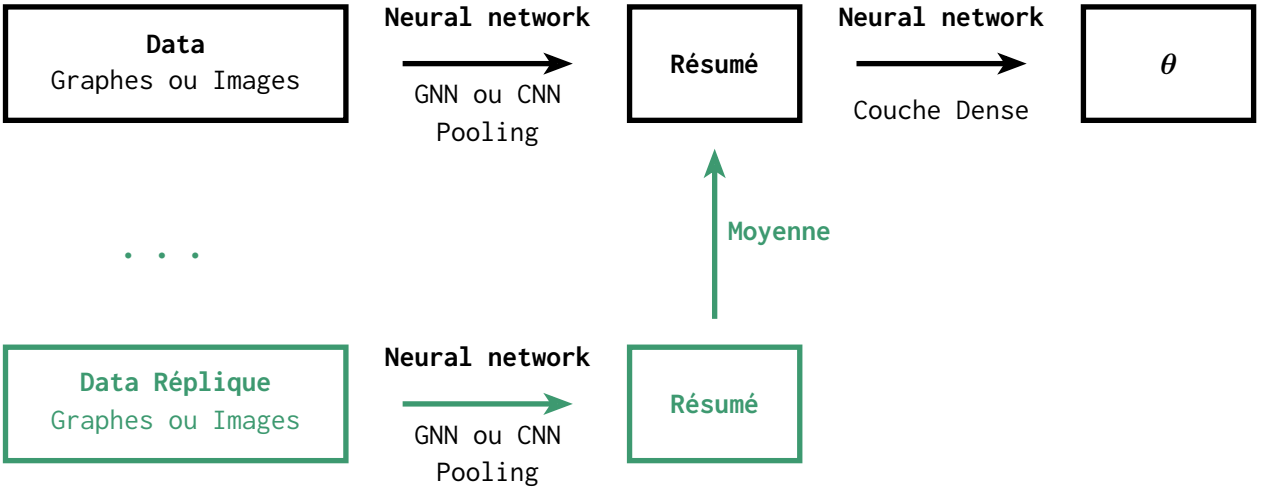
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1 Détails méthodes

Deepset

Les données répliquées peuvent (générées sur un même θ) peuvent mener à du biais.

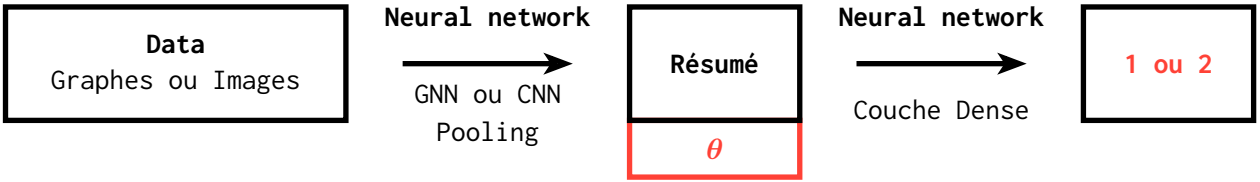


Neural Likelihood Suface (NLS)

Il a été montré que la vraisemblance peut être approchée de la façon suivante :

$$L(\theta|y) = \frac{h(y,\theta)}{1-h(y,\theta)}$$

avec $h(y,\theta)$ un classifier qui associe $C=1$ si la réalisation y et le paramètre θ sont dépendant, $C=2$ si indépendants.



Ce classifier permet de trouver la vraisemblance pour un couple (donnée, θ). Pour une donnée, le meilleur θ est trouvé par gridsearch ou optimisation.

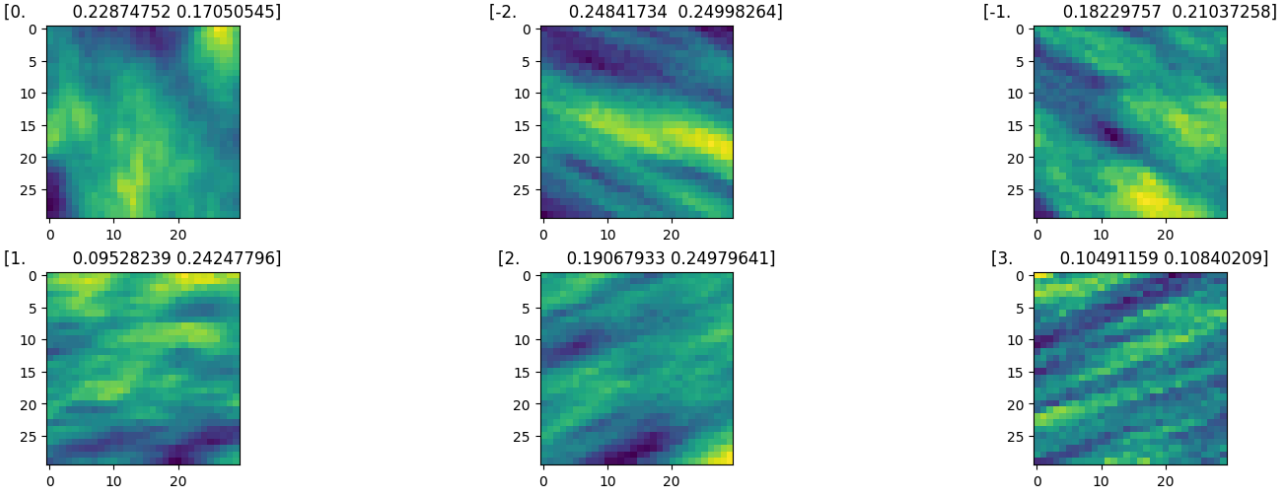
Baseline

1. θ_{train} la moyenne de tout les θ utilisés dans le train.
2. On regarde le MAE de chaque θ du train avec ce θ_{train}
3. La baseline est la moyenne de ces MAE

2 Résultats

Exemple images

1000 θ (advection, range time, range space) de train et 100 θ de test,
5 images 30×30 par θ.



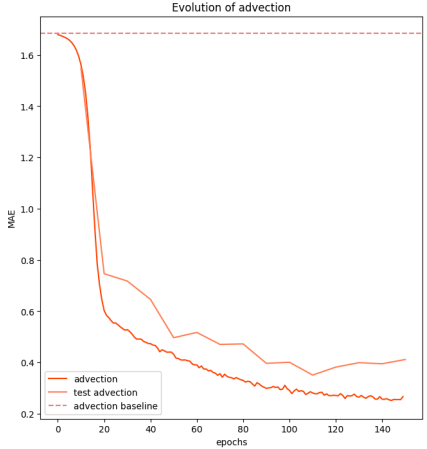
Méthodes

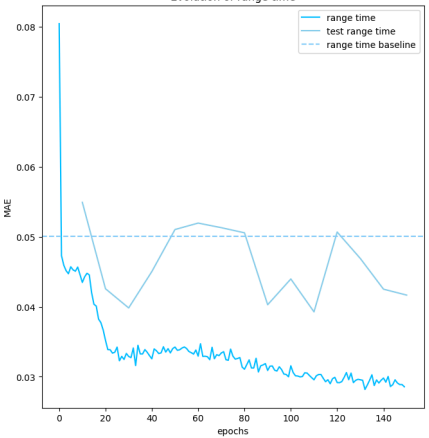
- 2D CNN + Dense
- 1D CNN + 1D CNN + Dense
- 1D CNN + LSTM + Dense
- …
- MLE

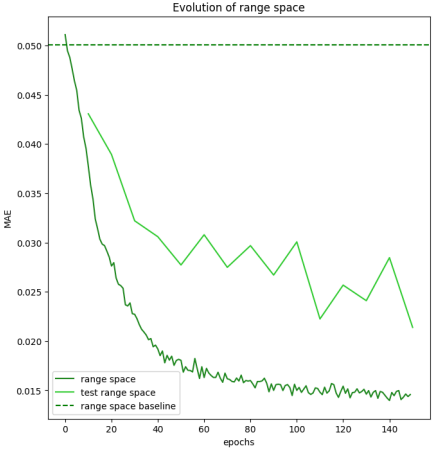
Les méthodes peuvent aussi être faites par NLS (sauf MLE)

Évolution loss

2.c.a 2D CNN + Dense

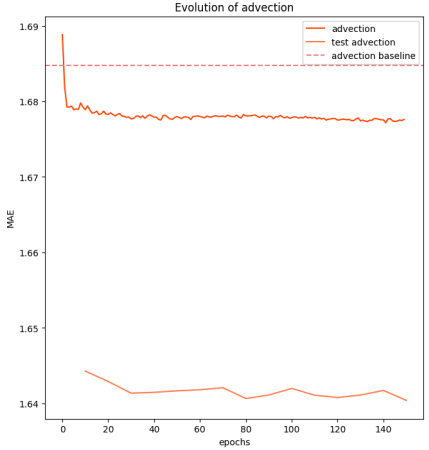


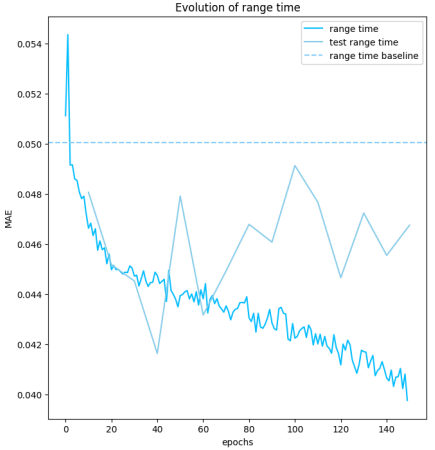


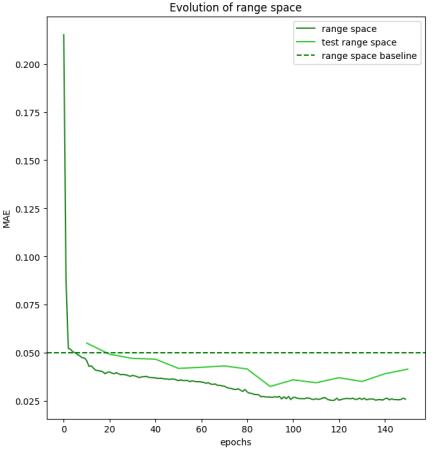


Évolution loss (ii)

2.c.b 1D CNN + 1D CNN + Dense

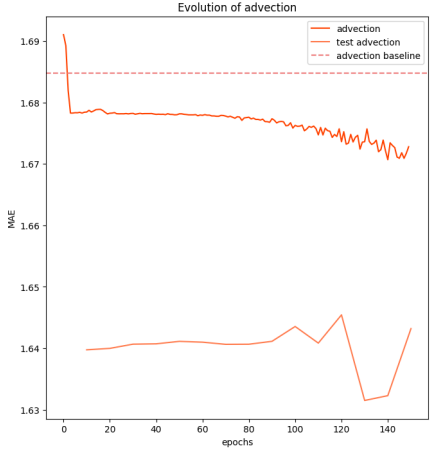


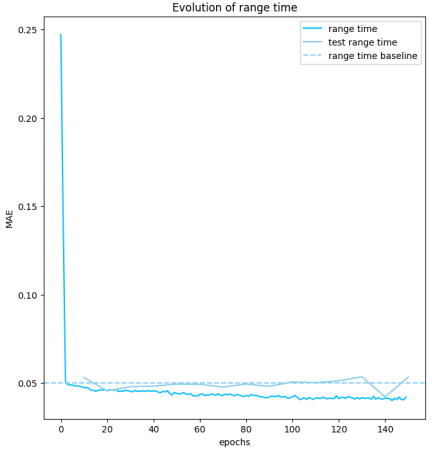


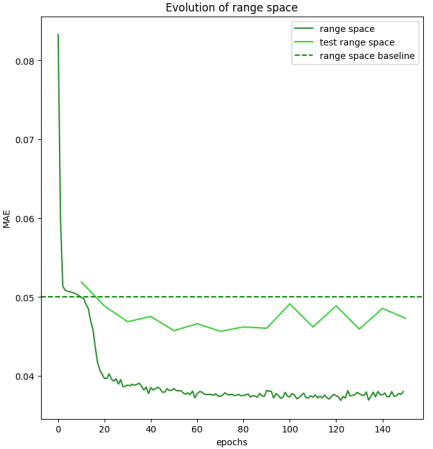


Évolution loss (iii)

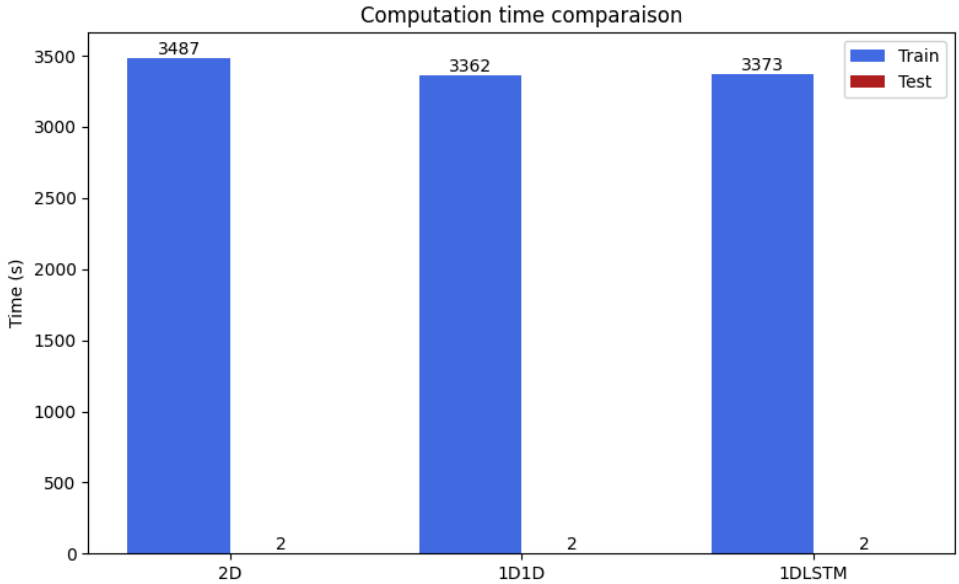
2.c.c 1D CNN + LSTM + Dense



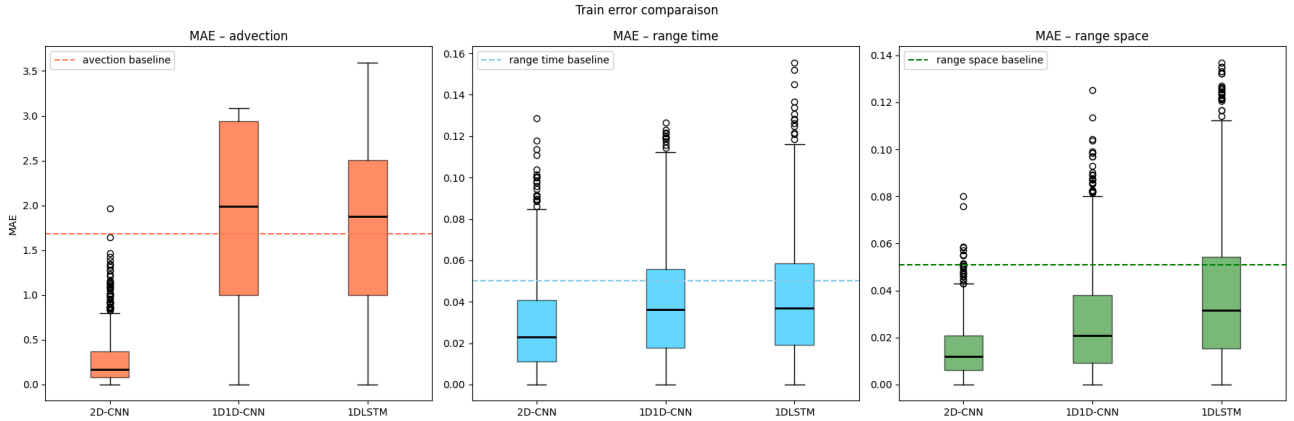




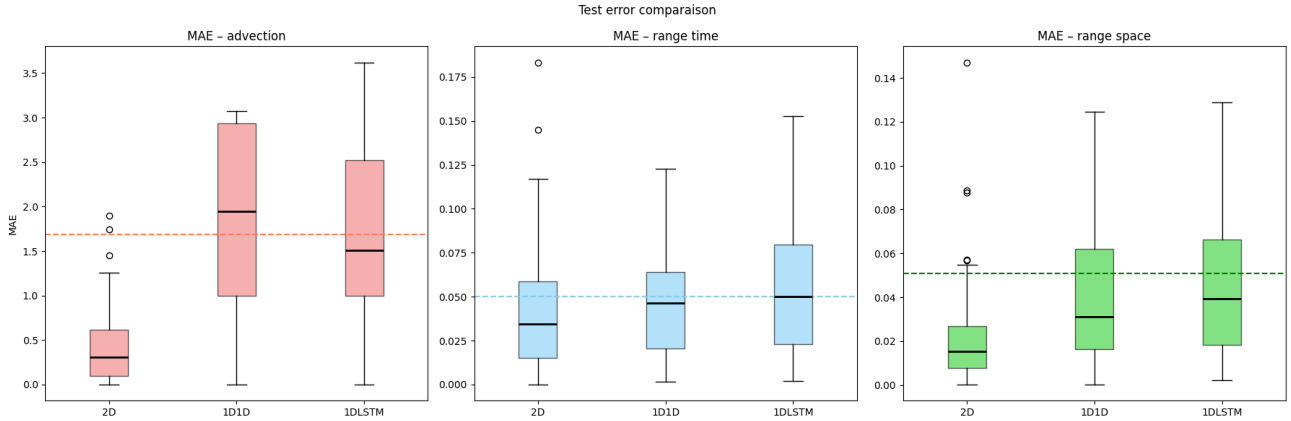
Graphes


Computation time comparaison
Time (s)
3500
3000
2500
2000
1500
1000
500
0
3487
2
3362
2
3373
2
Train
Test
2D
1D1D
1DLSTM

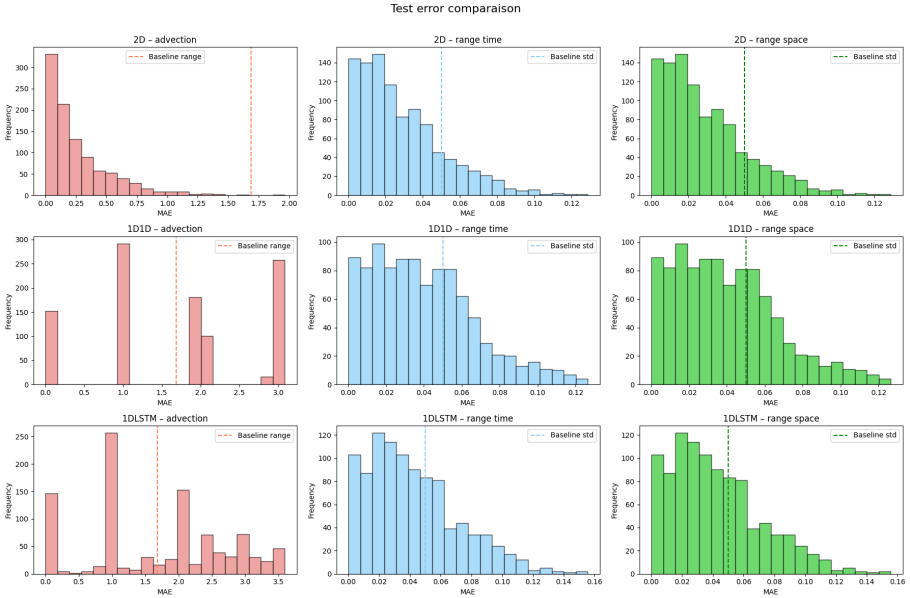
Graphes (ii)


Train error comparaison
MAE – advection
avection baseline
MAE
2D-CNN
1D1D-CNN
1DLSTM
MAE – range time
range time baseline
MAE – range space
range space baseline

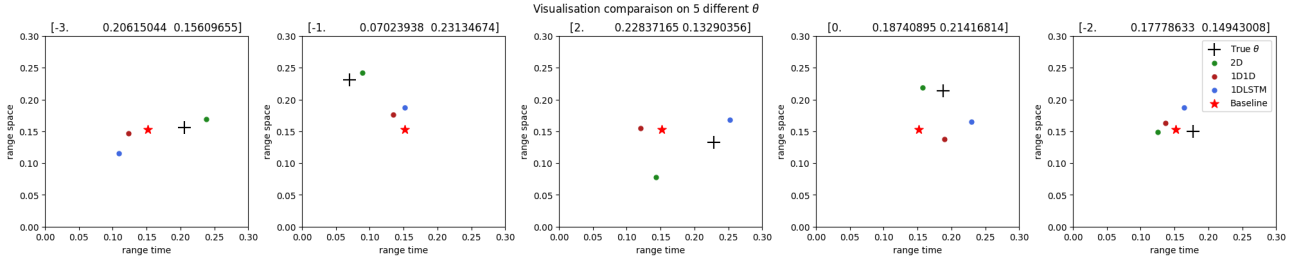
Graphes (iii)


Test error comparaison
MAE – advection
MAE – range time
MAE – range space
MAE
2D
1D1D
1DLSTM

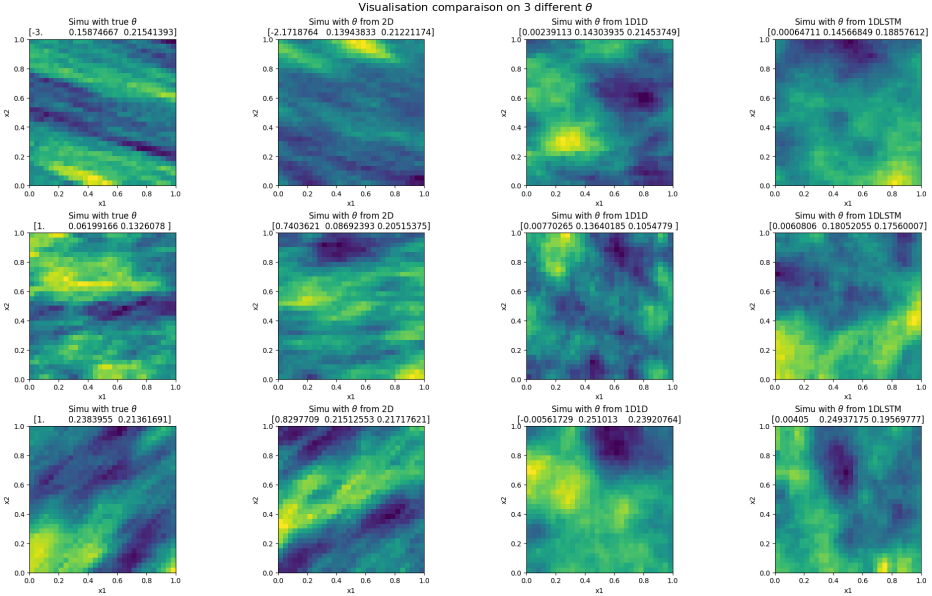
Graphes (iv)


Test error comparaison
2D – advection
2D – range time
2D – range space
1D1D – advection
1D1D – range time
1D1D – range space
1DLSTM – advection
1DLSTM – range time
1DLSTM – range space
Baseline range
Baseline std
Frequency
MAE

Graphes (v)


Visualisation comparaison on 5 different θ
[-3. 0.20615044 0.15609655]
[-1. 0.07023938 0.23134674]
[2. 0.22837165 0.13290356]
[0. 0.18740895 0.21416814]
[-2. 0.17778633 0.14943008]
range space
range time
True θ
2D
1D1D
1DLSTM
Baseline

Graphes (vi)


Visualisation comparaison on 3 different θ
Simu with true θ
[-3. 0.15874667 0.21541393]
Simu with θ from 2D
[-2.1718764 0.13943833 0.21221174]
Simu with θ from 1D1D
[0.00239113 0.14303935 0.21453749]
Simu with θ from 1DLSTM
[0.00064711 0.14566849 0.18857612]
Simu with true θ
[1. 0.06199166 0.1326078]
Simu with θ from 2D
[0.7403621 0.08692393 0.20515375]
Simu with θ from 1D1D
[0.00770265 0.13640185 0.1054779]
Simu with θ from 1DLSTM
[0.0060806 0.18052055 0.17560007]
Simu with true θ
[1. 0.2383955 0.21361691]
Simu with θ from 2D
[0.8297709 0.21512553 0.21717621]
Simu with θ from 1D1D
[-0.00561729 0.251013 0.23920764]
Simu with θ from 1DLSTM
[0.00405 0.24937175 0.19569777]
x1
x2

MLE

Pour 20 (et non 100) images 30x30 :

```
Time to test : 1400.25s
Test loss : [0.38284003 0.07327259 0.02481868] ± [0.89858435 0.06108345 0.0316696 ]
```

3 Simulation Caroline sans nugget

Code simulation spatio-temporel

Code reçu avant les vacances

Fonctionne

Voir paramètres à trouver :

- std ?
- range ?
- pas de temps ?

Ajouter advection ?

4 Administration 🔥

Grenoble

Hôtel : 6 rue crépu Grenoble

Train Paris-Grenoble : mardi 2 18h11, 103€

Inscription

Inscrit officiellement

Côté Mines :

- Convention d’acceuil
- Remboursement Navigo